

■ BUILD MANIFEST

WMS Roadmap

Cover the basics of every module first, then come back and deepen each one — rather than gold-plating one module before the rest exist at all. The last two sessions ran long on a Plan View / 3D-camera / physical-model detour (Ground & Floor Putaway, depth-tier stacking, a real camera-zoom bug, genuine bin numbering) before closing it out and doing a daily code-health pass.

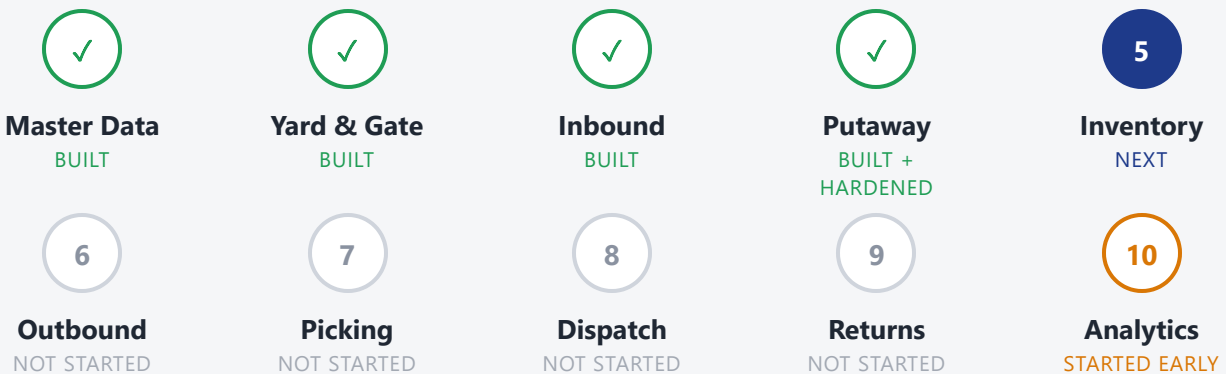
Repo **VasuRao97/WMS**

Updated **2026-09-07**

Latest commit **10ad14da**

Module build order

4 of 10 shipped · Analytics started early, out of sequence



What's landed since the last update

2 sessions' worth

Ground/Floor Putaway — real logic built end to end

BUILT

Its own independent `suggestGroundBin()` (not shared with Rack's own — a deliberate ask: "keep all logic for different storage type separate"), a column-lifecycle rule that closes a column to new putaway once picking starts on it (feeding the still-unbuilt reslotting engine), and per-class `respectsColumnBoundaries A/B/C/D` toggles. Also closed a real, unrelated gap along the way — `maxSkusClassA/B/C` had been a completely dead field since day one; now has a real Company Settings editor covering both Rack and Ground/Floor.

Ground/Floor: back-to-back depth-tier stacking

BUILT

A real bin can now be followed by more bins stacked back-to-back going further from the aisle, on ONE side of it — a genuine new physical-model capability (`Location.depthTier`), not a rendering fix.

Locations generator simplified

BUILT

The Second Range/Mirror UI is gone entirely, for both Rack and Ground — every `generate()` call now always builds exactly one flank, so ops staff get one simple instruction instead of a mirroring concept to learn.

3D Plan View: real camera-zoom bug fixed

BUILT

Checking one aisle vs. all of them used to produce the identical zoom level whenever row-depth dominated aisle-width — the ordinary case. Replaced with a real bounding-sphere fit so every axis genuinely contributes.

3D Plan View: genuine per-row bin numbering

BUILT

A crisp (WebGL, not blurry DOM) number now runs 1..N down each detailed aisle's own row, so it's unambiguous where a lane starts and ends. Caught and fixed a real follow-up bug the same day — numbers could land inside a neighboring aisle's own boxes once the gap between aisles got small.

A real flankNumber collision bug, plus a hard guard rail

BUILT

Two unrelated storage types generated onto the same Aisle number could silently share/corrupt flank identities. Fixed the allocator, backfilled one real warehouse's corrupted rows (client's go-ahead), and added a hard block against superimposing conflicting storage-type families on one Aisle going forward.

Ground/Floor box-size parity in the Plan View

BUILT

A 4×4 Ground bin now genuinely renders as a 4-row×4-box grid — 16 individually visible, individually clickable positions — in both 2D and 3D, instead of one stretched or width-scaled box.

Putaway Simulation: Ground/Floor support

BUILT

Closes the last gap in the sandbox's storage-type coverage. Investigating it surfaced and fixed two real, pre-existing Plan View bugs left over from the original Ground rewrite — both live in the shared components, so the real Locations page benefits too.

"Rows 1-N" row-position summary

BUILT

Each flank's callout now shows a running position count, matching the same "paired by position, not raw stored number" convention the rest of the Plan View already uses.

ProductCategory global-pollution cleanup, plus a standing convention

BUILT

~13 fake-looking categories had leaked into the real Category dropdown from roughly a dozen past throwaway-test sessions, since ProductCategory has no companyId — it's genuinely shared platform-wide. Traced and removed every one (deleting confirmed-automated test companies outright; re-pointing real records off the rest without touching them), verified live down to a single "Uncategorized" entry. A new standing rule now requires any future test cleanup to also delete its own category rows, not just the company — plus a same-day lint-driven pass across the backend that found and fixed a handful of genuinely unnecessary type assertions, zero behavior change.

Immediate candidates — next session

pick one

YOUR OWN PICK

FMS × ABC combined classification, for exact material putaway

CONFIRMED NEXT

"go in depth for FMS model of inventory as well, looking at the combo of abc and fms we need to make a logic of exact material putaway." FMS (Fast/Medium/Slow-moving, ranked by movement *frequency*) is a different axis than ABC (ranked by dispatched value) — a combined matrix is the natural next refinement on top of the dock-relative placement work already built. Nothing designed yet — real open questions (what drives FMS, how many tiers, how the matrix actually changes bin suggestion) are ready to work through.

01

Reslotting/consolidation engine

The daily suggestion engine flagged for a while now — which SKUs are misplaced, which bins/pallets can be consolidated. Schema's ready and genuinely unblocked, but needs its own design pass first.

needs design pass

02

Inventory (on-hand view)

Next in the stated build order. There is still no screen anywhere showing "what's on hand at Location X," even though real ledger data exists now. Comparatively light — mostly a read/report over data already being written.

03

Outbound order maker

The next real module after Inventory — destination + vehicle capacity checks (weight and volume), triggering a pick list. Benefits from Inventory existing first.

04

Dock-out → Gate-out signal

A notification once documentation/paperwork time after dock-out is up, before Gate Out is actually allowed. Needs its own design conversation, same care as Dock Scheduling.

needs design pass

05

Dock + Staging + Yard visualizer

Paused mid-conversation with open questions still unanswered (positional data on Dock Doors/Yard Slots, live vs. static). Needs a spatial-data pass before anything else.

needs design pass

Deferred — lower priority

don't build unprompted

Self-service driver check-in

PARKED

Schema-only (SelfCheckInRequest) — still needs its own align-before-coding pass; drivers have no login/User account anywhere in this system.

Real GS1 / unique-barcode parsing ("Reading B")

PARKED

The tyres/FMCG-case problem. Supervisor-approve fallback makes those categories usable without it today.

Camera-based scanning

PARKED

Hardware-scanner (keyboard-wedge) input works everywhere today; no camera/decode library wired up.

Real SMS / Email / WhatsApp / Voice providers

PARKED

Every channel is still a stub that only logs. MSG91 and Exotel were the research leads — neither chosen or wired up.

Dock Scheduling

PARKED

Zero design/schema. Only its future output (a dock number, a driver's assigned dock) has anywhere to land today.

Stillage's own Putaway logic, and ASRS's own strategy

PARKED

Rack and Ground/Floor both now have real, physically-correct bin-suggestion logic. Stillage still inherits nothing (always NEEDS_BIN); ASRS still inherits SPR's generic behavior — "let's finish racked and ground first."

No cancel/undo paths yet

PARKED

A mistaken Gate In, an un-completable Putaway task, and correcting an already-completed mis-putaway all have no correction flow today.

Batch/Lot + expiry capture, and closing a short receipt

PARKED

Both resurfaced twice in competitor research (feeds a future FEFO policy) — never picked up.

MHE travel-time-based placement

PARKED

DockLocationDistance (real meters) and Equipment loaded/unloaded speed are schema-only — no data-entry tooling, no consuming logic. Today's placement uses a flat Level rule instead.

Blacklist enforcement, Yard analytics dashboard, WMS/TMS integration

PARKED

Explicitly parked — don't propose as quick wins. (Distinct from the real Analytics module now underway.)

Platform & ops: SUPER_ADMIN creation, permission matrix, invite-link flow, Inventory Control Policy, Opening Balance, dispatch-proximity calc, real SAP/ERP integration, cloud deployment

PARKED

All fully deferred — no schema or design work done.

How to use this file: at the end of a session, update the module status and the immediate-candidates list before committing — that's what keeps this useful for picking up work cold, instead of re-deriving it from CLAUDE.md's full history each time.